

ORGANIC MOLECULES

Module map

- Module 03: Organic Molecules
- Connected lab: lab-03_microscopy.md
- Use this deck as the visual path through the module's evidence and retrieval work.

Module 03

Organic Molecules

1

Carbon
skeletons

2

Building
and
breaking
polymers

3

Carbohydrates
and lipids

4

Proteins
and shape

5

Nucleic
acids and
information

Lab evidence

lab-03_microscopy.md

ORGANIC MOLECULES

Learning objectives

- Explain why carbon is central to biological molecule diversity.
- Distinguish monomers, polymers, dehydration synthesis, and hydrolysis.
- Compare the functions of carbohydrates and lipids.
- Connect protein shape to protein function and denaturation.
- Describe how nucleic acids encode biological information.

OBJECTIVE 1

Explain why carbon is central to biological molecule diversity.

OBJECTIVE 2

Distinguish monomers, polymers, dehydration synthesis, and hydrolysis.

OBJECTIVE 3

Compare the functions of carbohydrates and lipids.

OBJECTIVE 4

Connect protein shape to protein function and denaturation.

OBJECTIVE 5

Describe how nucleic acids encode biological information.

ORGANIC MOLECULES

Topic sequence

- Carbon skeletons: Carbon forms diverse skeletons that make biological molecules possible.
- Building and breaking polymers: Dehydration synthesis builds polymers, and hydrolysis breaks them down.
- Carbohydrates and lipids: Carbohydrates and lipids store energy and provide structure in different ways.
- Proteins and shape: Protein function depends on amino acid sequence and three-dimensional shape.
- Nucleic acids and information: DNA and RNA store and move information using nucleotide sequences.

01

Carbon skeletons

Carbon forms diverse skeletons that make biological molecules possible.

02

Building and breaking polymers

Dehydration synthesis builds polymers, and hydrolysis breaks them down.

03

Carbohydrates and lipids

Carbohydrates and lipids store energy and provide structure in different ways.

04

Proteins and shape

Protein function depends on amino acid sequence and three-dimensional shape.

05

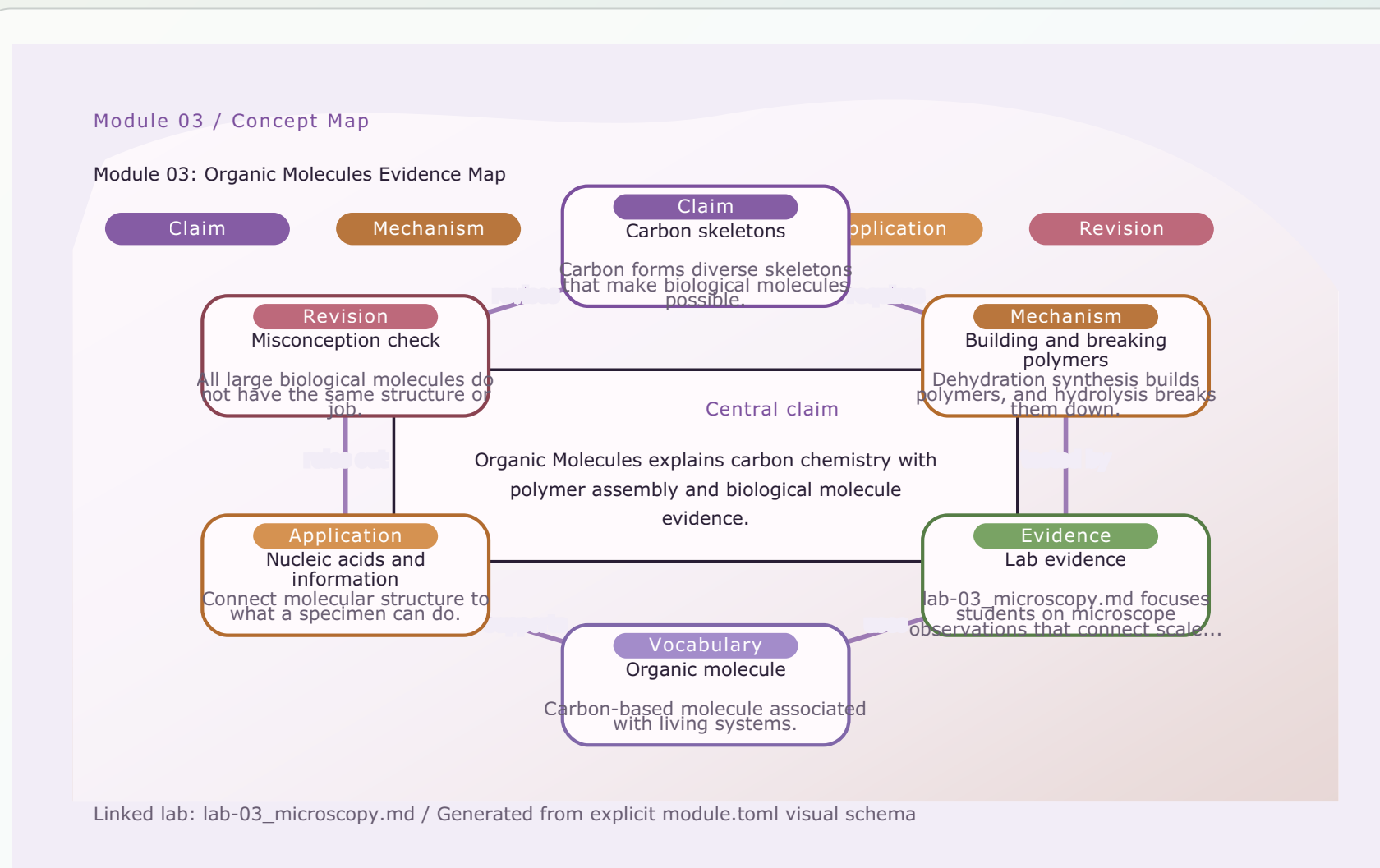
Nucleic acids and information

DNA and RNA store and move information using nucleotide sequences.

ORGANIC MOLECULES

Module 03: Organic Molecules Evidence Map

- Module focus: Organic Molecules
- Central claim: Organic Molecules explains carbon chemistry with polymer assembly and biological molecule evidence.
- Key nodes to connect: Carbon skeletons, Building and breaking polymers, Lab evidence
- Reasoning move: use 'requires' to explain relationships, not memorize terms.
- Lab connection: lab-03_microscopy.md supplies an evidence surface for the map.



Teaching note: Ask students to explain one edge in their own words before moving on.

ORGANIC MOLECULES

Module 03: Organic Molecules Reasoning Sequence

- Module focus: Organic Molecules
- Trace the model: Carbon skeletons -> Building and breaking polymers -> Carbohydrates and lipids -> Proteins and shape
- Inputs become outputs: Carbon skeletons, Building and breaking polymers -> Explain why carbon is central to biological molecule diversity., Distinguish monomers, polymers, dehydration synthesis, and hydrolysis.
- Feedback check: lab-03_microscopy.md evidence checks whether students can explain carbohydrates and lipids.
- Lab connection: lab-03_microscopy.md gives students a process to observe or test.

Module 03 / Process Model

Module 03: Organic Molecules Reasoning Sequence

Inputs

- Carbon skeletons
- Building and breaking polymers
- Evidence from lab-03_microscopy.md

Outputs

- Explain why carbon is central to biological molecule diversity.
- Distinguish monomers, polymers, dehydration synthesis, and...
- Describe how nucleic acids encode biological information.

Reasoning sequence

1

Carbon skeletons

Carbon forms diverse skeletons that make biological molecules possible.

2

Building and breaking polymers

Dehydration synthesis builds polymers, and hydrolysis breaks them down.

3

Carbohydrates and lipids

Carbohydrates and lipids store energy and provide structural support in different ways.

4

Proteins and shape

Protein function depends on amino acid sequence and three-dimensional shape.

5

Nucleic acids and information

DNA and RNA store and move information using nucleotide sequences.

Feedback

lab-03_microscopy.md evidence checks whether students can explain carbohydrates and lipids.; Retrieval questions reveal whether terms are...

Constraint

All large biological molecules do not have the same structure or job.; DNA and RNA store and move information using nucleotide sequences.

Linked lab: lab-03_microscopy.md

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Terms as evidence handles

- Organic molecule: Carbon-based molecule associated with living systems.
- Monomer: Small building block that can join into a polymer.
- Polymer: Large molecule built from repeating subunits.
- Carbohydrate: Sugar or sugar polymer used for energy or structure.
- Lipid: Hydrophobic molecule including fats, phospholipids, and steroids.
- Protein: Polymer of amino acids that performs cellular work.

ORGANIC MOLECULE

Carbon-based molecule associated with living systems.

MONOMER

Small building block that can join into a polymer.

POLYMER

Large molecule built from repeating subunits.

CARBOHYDRATE

Sugar or sugar polymer used for energy or structure.

LIPID

Hydrophobic molecule including fats, phospholipids, and steroids.

PROTEIN

Polymer of amino acids that performs cellular work.

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Lab connection

- Lab file: lab-03_microscopy.md
- Lab evidence should test or illustrate:
Carbohydrates and lipids
- Students should leave with a concrete observation, comparison, or model tied to the module claim.

QUESTION

Carbon skeletons

EVIDENCE

lab-03_microscopy.md

CLAIM

Explain why carbon is central to biological molecule diversity.

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Contrast check

- Do not stop at: Carbon forms diverse skeletons that make biological molecules possible.
- Stronger explanation: DNA and RNA store and move information using nucleotide sequences.
- The goal is to move from a named idea to a testable biological explanation.

SURFACE

Carbon forms diverse skeletons that make biological molecules possible.

DEEPER

DNA and RNA store and move information using nucleotide sequences.

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Module 03: Organic Molecules Retrieval and Lab Check

- Module focus: Organic Molecules
- Retrieval prompt: Why can carbon make many different biological molecules?
- Answer check: Explain why carbon is central to biological molecule diversity.
- Required terms: Organic molecule, Monomer, Polymer
- Lab connection: lab-03_microscopy.md; lab-03_microscopy.md supplies evidence for microscope observations that connect scale, structure, and biological function.

Module 03 / Retrieval Card

Module 03: Organic Molecules Retrieval
and Lab Check

Routine

Cover notes -> answer aloud
-> cite evidence -> revise

1 Cover notes

2 Answer aloud

3 Cite evidence

4 Revise

Why can carbon make many different biological molecules?

Check: Explain why carbon is central to biological molecule diversity.

How are dehydration synthesis and hydrolysis opposite processes?

Check: Distinguish monomers, polymers, dehydration synthesis, and hydrolysis.

What distinguishes a monomer from a polymer?

Check: Compare the functions of carbohydrates and lipids.

How do carbohydrates and lipids differ in structure and function?

Check: Connect protein shape to protein function and denaturation.

Terms to use

Organic molecule, Monomer, Polymer, Carbohydrate, Lipid

Lab connection

lab-03_microscopy.md supplies evidence for microscope observations that connect scale, structure, and biological function.

Intro biology routine: claim, evidence, reasoning, revision.

Teaching note: Pause here. Students answer first without notes, then compare to the checks.

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Practice quiz bridge

- 1. Why is carbon the backbone of most biological molecules?
- 2. Cells join two monomers into a polymer by removing a water molecule. This reaction is called:
- 3. A protein is heated until it loses its three-dimensional shape and stops working. This change is called:
- 4. Which pairing of molecule and job is correct?

QUESTION 1**Key: C**

Why is carbon the backbone of most biological molecules?

QUESTION 2**Key: A**

Cells join two monomers into a polymer by removing a water molecule. This reaction is called:

QUESTION 3**Key: D**

A protein is heated until it loses its three-dimensional shape and stops working. This change is called:

QUESTION 4**Key: B**

Which pairing of molecule and job is correct?

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Synthesis and exit ticket

- Exit claim: explain carbon skeletons using evidence from lab-03_microscopy.md.
- Must include: Organic molecule, Monomer, Polymer.
- Revision prompt: what evidence would change your explanation?

CLAIM

Carbon skeletons

EVIDENCE

lab-03_microscopy.md

REVISION

What would change your mind?